Gmail **Peter Crimson <crimsonpeter317@gmail.com>**

Guidance on Flag Adjudication and Temporal Analysis

3 messages

Peter Crimson <crimsonpeter317@gmail.com> Sat, Sep 5, 2026 at 9:13 PM
To: "lifenglin@arizona.edu" <lifenglin@arizona.edu>
Cc: shiyinli@arizona.edu, "bulunte@arizona.edu" <bulunte@arizona.edu>

Good evening Dr. Lin,

Please excuse me for sending this email from an external Gmail account. It seems that my UA Outlook account is experiencing email delivery issues. I am currently unable to send emails to internal UA.edu accounts, although I can still communicate with external accounts such as Gmail and D2L. I also discovered that the email I sent yesterday was not delivered successfully. Here is the original email.

I would appreciate your guidance on three issues arising from our review of the AI validation results.

1. Protocol-related discrepancies. Shiyin noted that differences between protocols and submitted manuscripts or supplements may reflect necessary changes during trial conduct. Should these flags therefore be excluded? My view is that, if we position agentic AI as a general tool for detecting reporting issues, such discrepancies remain useful candidates for human review. We could distinguish justified changes from reporting problems rather than exclude protocol-related flags as a group. To characterize the tool's capabilities, I propose focusing primarily on descriptive and exploratory analyses of the types of issues it identifies.

2. The secondary temporal comparison. We previously planned to compare pre- and post-pandemic flag counts to explore whether the wider availability of AI tools might be associated with improved reporting. However, protocol-related discrepancies may reflect practical trial constraints rather than reporting quality, potentially confounding the interpretation of temporal differences. For this analysis, I suggest focusing on statistical reporting errors and inconsistencies within the submitted manuscript and non-protocol supplements. This narrower scope may reduce that concern.

3. Proposed analysis. Applying the scope described above, preliminary screening leaves 287 of the original 412 flags after excluding protocol-versus-submitted-document and protocol-versus-protocol issues, before any rebuttal or human adjudication. To examine differences across years, I propose constructing count outcomes from this subset, including internal reporting inconsistencies, statistical reporting errors, and discrepancies in within-document totals. We could conduct an exploratory analysis using a hierarchical Bayesian count model or a GLM with standard errors clustered by paper, depending on how the observations are structured.

Would you consider this distinction between the tool's broader review scope and the narrower outcomes for temporal analysis appropriate? I would also welcome your advice on the proposed modeling approach and the bootstrap options for inference that we previously discussed.

Best regards,
Bulun Te

Lin, Lifeng - (lifenglin) <lifenglin@arizona.edu> Sat, Sep 5, 2026 at 10:49 PM
To: Peter Crimson <crimsonpeter317@gmail.com>
Cc: "Li, Fiona - (shiyinli)" <shiyinli@arizona.edu>, "Te, Bulun - (bulunte)" <bulunte@arizona.edu>

No worries, but you might want to figure out this issue in case you miss any important messages from others.

Below is my reply.

“Hi Bulun,

Thanks. I agree with the general direction. A few thoughts:

1. I have also noticed, when using AI to assist with reviewing meta-analysis papers, that AI is very good at identifying discrepancies between protocols and final manuscripts. Such discrepancies are not necessarily problematic, and many reporting guidelines allow changes as long as they are documented and justified. However, in my experience, authors rarely provide such documentation. Since this issue appears to be quite prevalent, I suggest that we retain it in our paper. For example, we could include a supplementary table summarizing discrepancies between protocols and final publications and add a subsection in the Results describing these findings. That said, I would view these discrepancies as somewhat less important than errors in the reporting of statistical results, which could directly affect downstream analyses, e.g.,, if the reported data are later used in meta-analyses.
2. I still think a temporal comparison could be interesting, particularly to explore possible changes around COVID-19 and the development and broader availability of LLMs. For this comparison, however, I agree that we should focus on more substantive statistical reporting errors and internal inconsistencies, rather than protocol-related discrepancies.
3. Your proposed modeling approach sounds reasonable. One consideration is that we should examine whether the assumptions underlying the parametric models are appropriate for the data. I initially thought we might use a nonparametric approach, such as nonparametric bootstrapping, although that may have limited power depending on the structure and sample size. I am open to parametric approaches if their assumptions appear reasonable.

Best,
Lifeng

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From: Peter Crimson <crimsonpeter317@gmail.com>
Date: Saturday, September 5, 2026 at 9:14 PM
To: Lin, Lifeng - (lifenglin) <lifenglin@arizona.edu>
Cc: Li, Fiona - (shiyinli) <shiyinli@arizona.edu>, Te, Bulun - (bulunte) <bulunte@arizona.edu>
Subject: [EXT] Guidance on Flag Adjudication and Temporal Analysis

External Email

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Peter Crimson <crimsonpeter317@gmail.com> Sat, Sep 5, 2026 at 11:33 PM
To: "Lin, Lifeng - (lifenglin)" <lifenglin@arizona.edu>
Cc: "Li, Fiona - (shiyinli)" <shiyinli@arizona.edu>, "Te, Bulun - (bulunte)" <bulunte@arizona.edu>

Hi Dr.Lin,

Thank you for your guidance. We will continue the work based on your suggestions. Once the adjudication work is complete, I will explore nonparametric approaches and compare the results with those from the parametric models.

Best,

Bulun Te

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